🔑 **Answer Key — Teacher Copy · Lesson 9-3**

MSTAR PRACTICE WORKSHEET

Lesson 9-3: Write Equations to Represent Relationships Between Two Variables

Grade 6 Mathematics · Standard 6.AT.11

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

Recreation World charges $24.95 per month. What is the total cost of 2 months of membership?

A. $24.95

B. $26.95

C. $49.90

D. $74.85

Two months at $24.95 each is 2 × 24.95 = 49.90.

Question 2 SELECTED RESPONSE — CORRECT: A

In the equation c = 24.95m for the gym membership, what does m represent?

A. The number of months of membership

B. The total cost in dollars

C. The monthly fee of $24.95

D. The name of the gym

m stands for the quantity Miguel chooses — the number of months. The letter c stands for the total cost that results.

Question 3 SELECTED RESPONSE — CORRECT: A

A canoe rents for $15 per hour. Let h = hours and c = total cost. Which equation represents the relationship?

- (A) c = 15h
- (B) c = 15 + h
- (C) h = 15c
- (D) c = h − 15

> Each hour costs $15, so the total is 15 times the number of hours: c = 15h.

Question 4 SELECTED RESPONSE — CORRECT: C

Using c = 15h, what is the total cost of a 3-hour canoe rental?

- (A) $15
- (B) $18
- (C) $45
- (D) $75

> Substitute h = 3: c = 15 × 3 = 45.

Question 5 SELECTED RESPONSE — CORRECT: B

In the gym membership situation, which quantity is the DEPENDENT variable?

- (A) The number of months
- (B) The total cost
- (C) The monthly fee of $24.95
- (D) The number of gyms in town

> The total cost depends on the number of months of membership — Miguel chooses the months, and the cost responds.

Question 6 SELECTED RESPONSE — CORRECT: B

The membership table shows: 1 month, $24.95; 2 months, $49.90; 3 months, $74.85. By how much does the total cost increase each month?

A $24.05

B $24.95

C $25.00

D $49.90

Each row grows by the monthly fee: 49.90 – 24.95 = 24.95, and 74.85 – 49.90 = 24.95.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

A spin studio charges $7.50 for each class. Let n be the number of classes a member takes and let c be the total cost in dollars. Which equation represents the relationship?

(A) c = 7.5 + n

(B) n = 7.5c

(C) c = n ÷ 7.5

(D) c = 7.5n

The studio charges $7.50 for every class, so the total cost is the rate times the number of classes: c = 7.5n. The dependent variable, the cost, stands alone on the left, and the independent variable, the classes, is multiplied by the rate.

- **A:** This adds the price once rather than charging it per class. Each of the n classes costs $7.50.
- **B:** This makes the number of classes depend on the cost. The cost is what depends on the classes.
- **C:** This divides by the price. Each class ADDS $7.50, so multiply.

Part B — correct answer: D

Which reasoning justifies the equation chosen in Part A?

(A) The number of classes is added to the price of one class, so 4 classes cost 7.5 + 4 = $11.50.

(B) The number of classes depends on the cost, so the classes belong alone on the left side of the equation.

(C) The $7.50 is shared among the classes, so 4 classes cost 7.5 ÷ 4 = $1.875 each.

(D) The $7.50 is charged for every single class, so the total is 7.5 times the classes; 4 classes cost 7.5 × 4 = $30.

A rate applies to every class, so the operation is multiplication and the total for 4 classes is 7.5 × 4 = $30. Adding gives $11.50, which charges the fee only once. The cost depends on the classes, not the other way around, so the cost is the variable that stands alone.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, C, E

A community pool charges $9 for each hour of lane rental. Let h be the number of hours rented and let c be the total cost in dollars. Select ALL statements that correctly represent this relationship.

- **A** c = 9h
- **B** h = 9c
- **C** The total cost is 9 times the number of hours rented.
- **D** c = 9 + h
- **E** In the equation, h stands for the number of hours rented and c stands for the total cost in dollars.
- **F** In the equation, c stands for the $9 charged for each hour.

The $9 is charged for every hour, so the total cost is the rate times the hours: c = 9h, which also checks against 4 hours at 9 × 4 = $36. Writing h = 9c makes the hours depend on the cost, and c = 9 + h charges the fee only once. Defining the variables means saying what each letter counts: h counts hours and c counts dollars of total cost, while the 9 is the fixed rate and is not a variable.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.